TLF-35A

Improved PTH and PIMD Performance

AGC
Your Dreams, Our Challenge

Benefits

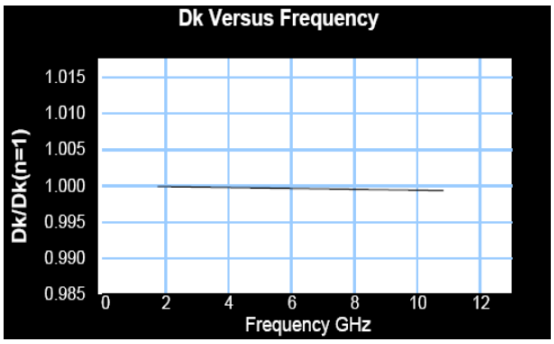
- Improved PIMD with DK3.5
- Improved PTH Quality
- Stable at high frequency
- Stable at high temp.
- Low moisture absorption
- Excellent Peel Strength
- Excellent price/performance Ratio

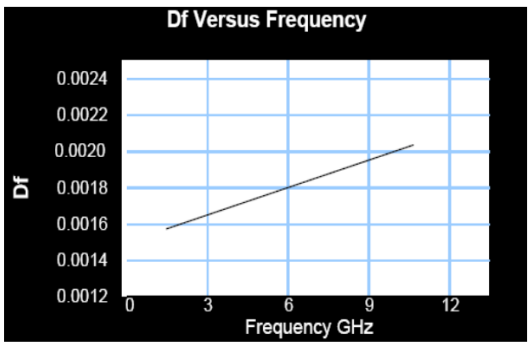
Applications

- Size effective Antenna
- Power Amplifiers
- LNA, Repeater PA
- Passive Components
- Filters / Couplers

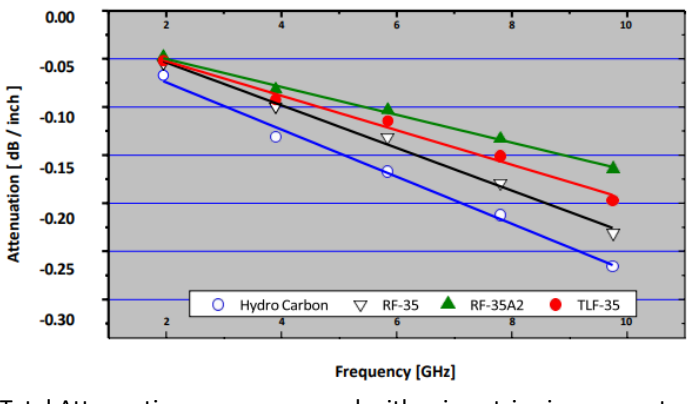
TLF-35A is an organic-ceramic laminate in Taconic's family of product. TLF-35A advanced is the best choice for low cost, high volume commercial microwave and radio frequency application. TLF-35A advanced has excellent peel strength for ½ ounce and 1 ounce copper and is designed to offer superior high frequency performance. Advanced TLF-35A laminates show similar electrical properties as TLF-35A but tighter DK tolerance.

TLF-35A advanced is designed to obtaining improved PIMD performances on size effective boards for antenna application. Most of sensitive PIMD required base material's Dielectric constant is around 3.0 whereas TLF-35A advanced laminates dielectric constant is 3.5 with similar PIMD levels.





Total Attenumation (Microstrip Resonator)



Total Attenuation were measured with microstrip ring resonator.
Material under test were 20mil dielectric thickness and 1 oz. copper.

Properties	Conditions	Typical Value	Unit	Test Method
Electrical Properties				
Dielectric Constant	@ 10 GHz	3.50 ± 0.05		IPC-TM 650 2.5.5.5.1 Mod
Dissipation Factor	@ 10 GHz	0.0026		IPC-TM 650 2.5.5.5.1 Mod
Surface Resistivity		3.0×10^9	Mohms	IPC-TM 650 2.5.17.1
Volume Resistivity		2.0×10^8	Mohms/cm	IPC-TM 650 2.5.17.1
Thermal Properties				
Thermal Conductivity		0.37	W/m/K	IPC-TM-650 2.4.50
CTE (50 to 150 °C)	X	9	ppm/°C	IPC-650 2.4.41
	Y	12		
	Z	80		
Mechanical Properties				
Peel Strength	1 oz. copper	1.8 (10)	N/mm (Lbs./linear in)	IPC-TM 650 2.4.8
Flexural Strength	Lengthwise	90 (13,000)	N/mm^2 (psi)	IPC-TM 650 2.4.4
	Crosswise	90 (13,000)	N/mm^2 (psi)	
Chemical / Physical Properties				
Flammability			V-0	UL-94
Water Absorption		0.03	%	IPC-TM 650 2.6.2.1

Typical Thicknesses[1]	
Inches	**mm**
0.030	0.76
0.060	1.52

Typical Panel Sizes[2]			
Inches	**mm**	**Inches**	**mm**
12 x 18	305 x 457	18 x 24	457 x 610
16 x 18	406 x 457	36 x 48	914 x 1,220

Available Copper Cladding						
Designation	**Weight**	**Copper Thickness**		**Rms Treated Side**		**Description**
CVH (CH)	½ oz./sq. ft.	~ .0007”	~ 18μm	19μin	0.48μm	Very low profile / Electrodeposited
CV1 (C1)	1 oz./sq. ft.	~ .0014”	~ 35μm	25μin	0.64μm	Very low profile / Electrodeposited
CLH	½ oz./sq. ft.	~ .0007”	~ 18μm	18μin	0.46μm	Reverse Treated / Electrodeposited
CL1	1 oz./sq. ft.	~ .0014”	~ 35μm	16μin	0.41μm	Reverse Treated / Electrodeposited
C2	2 oz./sq. ft.	~ .0028”	~ 70μm	27μin	0.69μm	Electrodeposited
CVH (CH)	½ oz./sq. ft.	~ .0007”	~ 18μm	19μin	0.48μm	Very low profile / Electrodeposited
CV1 (C1)	1 oz./sq. ft.	~ .0014”	~ 35μm	25μin	0.64μm	Very low profile / Electrodeposited

* All test data provided are typical values and not intended to be specification values. For review of critical specification tolerances, please contact a company representative directly.
* TLF-35A can be manufactured in increments of 0.030”(0.76mm).
* Standard panel size is 18” x 24” (457 mm x 610 mm).
* Please contact AGC for availability of additional thicknesses, other sizes & any other type of cladding.

agc-ml.multimaterial@agc.com